

Mofe Omolade

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EDUCATION

University of British Columbia, *BASc in Computer Engineering*

Expected May 2029

Relevant Coursework: Circuit Analysis, Computing Systems, Software Construction, Embedded C, Electronics Lab.

EXPERIENCE

FulcrumAir, *Electronics Engineering Intern*, Calgary, AB

May 2026 - Present

- Performed design and layout of an 8-layer motor control PCBA in Altium around Teensy 4.1, IFX007T, MAX33040E, and LMR50410-Q1, optimizing layer stackup, thermal regulation, and power integrity.
- Developed C++ firmware for motor control, parsing Hall sensor data at 1 kHz to achieve <0.2 mm position accuracy.
- Implemented a power hold-up circuit to enable interrupt-triggered EEPROM writes within <5 ms of power cutoff.
- Designed a 2-layer control breakout board supporting CAN and RS-485 buses for multi-actuator robotic systems, integrating 4 camera channels, multi-axis servo buses, limit switches, load cells, and a dedicated servo debug port.
- Integrated TPSM33625 and TPSM53604 ICs for regulated 12V and 5V supplies across sensor and camera rails.

University of Calgary, *Electrical Engineering Research Assistant*, Calgary, AB

July 2023 - August 2023

- Modelled and simulated circuits in LTspice to support research on novel Op-Amp-based oscillator topologies.
- Validated circuit performance, achieving >95% correlation between simulation and lab measurements.
- Utilized oscilloscopes, signal generators, and DMMs to debug test circuits and verify frequency responses.

UBC Rover, *Electrical Subteam*, Vancouver, BC

September 2025 - Present

- Designed power architecture, telemetry communication system, and motor drive boards for an autonomous test rover.
- Tested a powertrain prototype around the L298N full-bridge motor driver and Nucleo F446RE MCU.
- Developed low-level C firmware using STM32 HAL for real-time telemetry, command parsing, and Hall effect encoder motor feedback at 115200 baud UART.

PROJECTS

Self-Balancing Two-Wheel Robot

- Designing a two-wheeled self-balancing robot using an ESP32-C3, MPU-6050 IMU, and L298N motor driver.
- Developed C++ I2C drivers to parse IMU registers at 1kHz and apply complementary filtering, reducing drift by 85%.
- Implemented a closed-loop PID control system to drive dual DC motors to maintain dynamic equilibrium.

Speedometer Heads-Up Display

- Designed a low-latency heads-up display using an Arduino Nano, GY-NEO6MV2 module, and SPI OLED display.
- Wrote C++ firmware to parse 10 Hz NMEA data streams to compute velocity within 2% of OBD-II metrics.
- Integrated a USB-C-to-DIN step-down regulator to safely isolate vehicle transient spikes up to 18V.

SYNCHRONOUS 24V-5V BUCK CONVERTER

- Designed topology with dual N-channel MOSFETs, floating gate drives, and analog Type II loop compensation.
- Validated open-loop dynamics, small-signal Bode stability, and transient response in LTspice simulations.
- Performed 4-layer PCB layout in Altium, minimizing ground and switching loops to maintain signal integrity.

SKILLS

- Hardware: PCBA Design, Power Electronics, CAN, SPI, I2C, UART, RS-485, STM32, ESP32, Arduino.
- Tools: Altium, KiCAD, LTspice, MATLAB, Git, VS Code, STM32 CubeIDE.
- Languages: C/C++, Python.
- Lab: Oscilloscopes, Logic Analyzers, DMM, Function Generators, Soldering, Circuit Debugging.